

Build Sand Table Map

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1. Course Content

Learning Objectives

- Master the complete process of building a sand table grid map using slam_toolbox
- Learn to use the Publish Point tool to create waypoints, and use the 2D Pose Estimate tool for relocation
- Understand the basic principles of road network planning and annotation strategies for sand table maps

[!IMPORTANT]

- This section is a comprehensive course on sand table maps, introducing the complete process of building a sand table map. For detailed principles, parameter debugging, etc., refer to [04-OpenClaw Embodied AI Practice - OpenClaw SLAM Mapping Navigation] and [Road Network Planning Navigation 1-6] chapters.

2. Preparation

- Start the road network container

```
jetson@yahboom:~$ roadnet start
access control disabled, clients can connect from any host
jetson@yahboom:~$
```

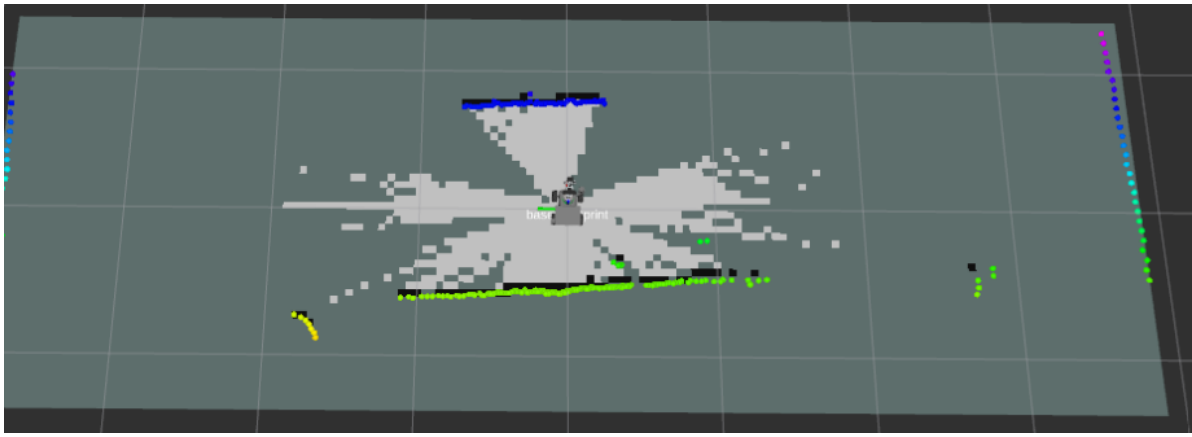
- Start the MicroROS chassis agent (skip if already started)

```
sh start_agent.sh
```

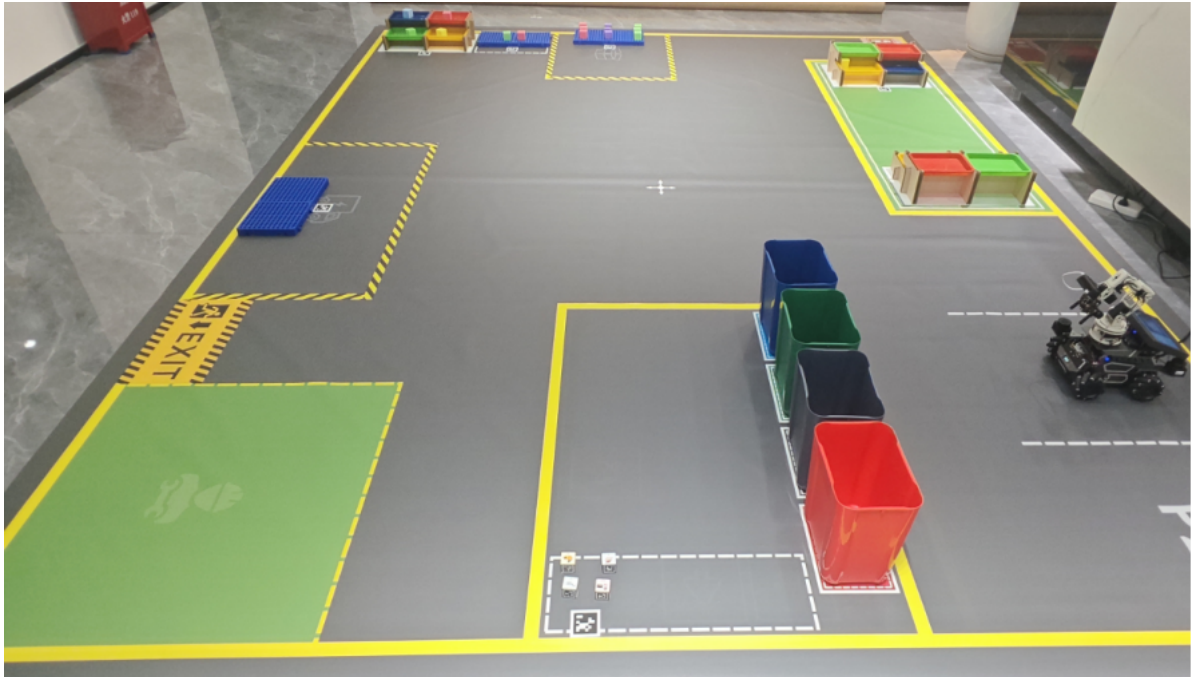
3. Build Grid Map

- Start SLAM mapping

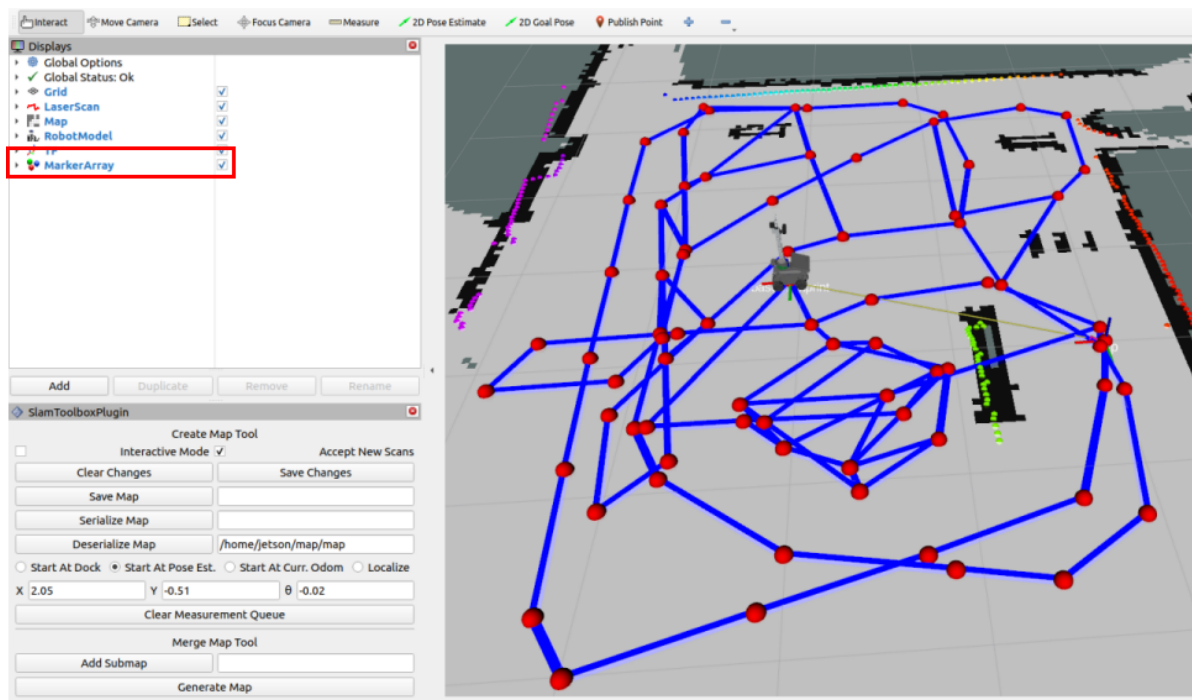
```
ros2 launch slam_mapping slam_toolbox.launch.py
```



- Control the robot to move and complete map construction



- **Check the number of pose points (important)**
 - Click the MarkerArray option in the Rviz Display panel to view the pose points of the pose map. Ensure there are enough pose points in the map area. If an area has too few or no pose points, the positioning effect in that area will be poor during navigation.



- Save the grid map

```
ros2 launch slam_mapping save_slamtoolbox_map.launch.py
```

```
jetson@yahboom: ~ 82x22
[map_saver_cli-1] map_saver lifecycle node launched.
[map_saver_cli-1] Waiting on external lifecycle transitions to activate
[map_saver_cli-1] See https://design.ros2.org/articles/node_lifecycle.html for more information.
[map_saver_cli-1] [INFO] [1763532975.329685271] [map_saver]: Creating
[map_saver_cli-1] [INFO] [1763532975.330069728] [map_saver]: Configuring
[map_saver_cli-1] [INFO] [1763532975.341867665] [map_saver]: Saving map from 'map'
[map_saver_cli-1] topic to '/home/jetson/map/map' file
[map_saver_cli-1] [WARN] [1763532975.359932772] [map_io]: Image format unspecified
[map_saver_cli-1] . Setting it to: pgm
[map_saver_cli-1] [INFO] [1763532975.361316136] [map_io]: Received a 224 X 598 map
[map_saver_cli-1] @ 0.05 m/pix
[map_saver_cli-1] [INFO] [1763532975.439369065] [map_io]: Writing map occupancy data
[map_saver_cli-1] to /home/jetson/map/map.pgm
[map_saver_cli-1] [INFO] [1763532975.443013992] [map_io]: Writing map metadata to
[map_saver_cli-1] /home/jetson/map/map.yaml
[map_saver_cli-1] [INFO] [1763532975.444847063] [map_io]: Map saved
[map_saver_cli-1] [INFO] [1763532975.446514786] [map_saver]: Map saved successfully
[map_saver_cli-1] [INFO] [1763532975.449017438] [map_saver]: Destroying
[map_saver_cli-1] [INFO] [map_saver_cli-1]: process has finished cleanly [pid 50218]
jetson@yahboom:~$
```

- Save the pose constraint map

```
ros2 launch slam_mapping save_slamtoolbox_map.launch.py map_type:=posegraph
```

```
jetson@yahboom:~$ ros2 launch m3_bringup save_map.launch.py map_name:=/home/jetson
/map/map map_type:=posegraph
[INFO] [launch]: All log files can be found below /home/jetson/.ros/log/2025-11-19
-14-19-41-297598-yahboom-66778
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [ros2-1]: process started with pid [67023]
[ros2-1] requester: making request: slam_toolbox.srv.SerializePoseGraph_Request(fi
lename='/home/jetson/map/map')
[ros2-1]
[ros2-1] response:
[ros2-1] slam_toolbox.srv.SerializePoseGraph_Response(result=0)
[ros2-1]
[INFO] [ros2-1]: process has finished cleanly [pid 67023]
jetson@yahboom:~$
```

4. Build Road Network Description File

- Start slam_toolbox localization and navigation (start on host)

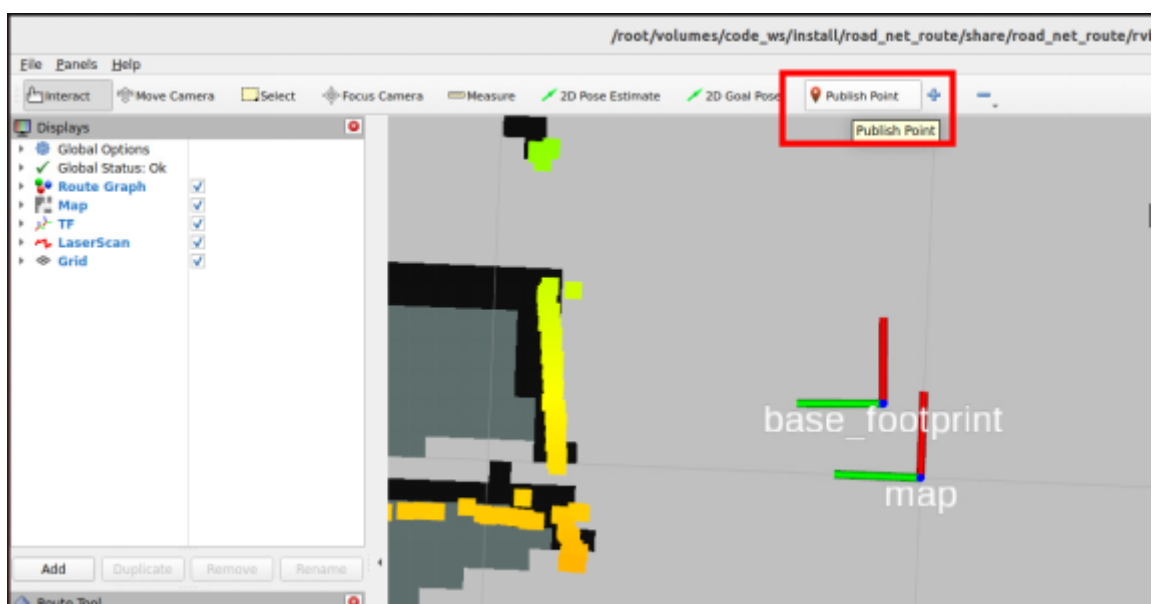
```
ros2 launch M3Pro_navigation toolbox_location_nav.launch.py gui:=False
```

- Annotate the road network file (start inside roadnet container)

```
roadnet bash
```

```
ros2 launch road_net_route route_tool.launch.py
```

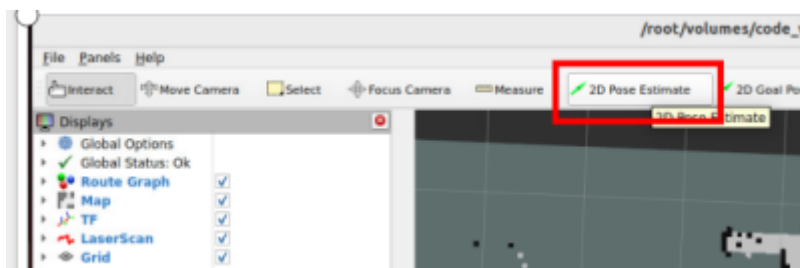
- Click the `Publish Point` tool in the rviz toolbar to enter waypoint selection mode



- Then click the left mouse button on the small blue dot at the center of the base_footprint coordinate system. The `position` field in the `Route Tool` tab on the lower left will automatically fill in the coordinates of the current point. Then click Create to create a waypoint.



- Move the robot to the next waypoint that needs to be marked, then click the **2D Pose Estimate** tool to give an estimated position near the approximate location of the robot for relocation.



- Repeat the above process until all required waypoints are marked and the road network is connected (annotate according to actual needs. For detailed marking process, refer to the [Road Network Planning Navigation] chapter tutorial).
- The reference sand table map road network is as follows:
- Here, waypoints are marked on the main roads and important areas of the sand table map. You can annotate the road network according to the tasks you actually need to perform.

